CM 615: Climate Change Impacts & Adaptation

Class Activity

Exploring Evidences of Climate Change over India Using Observations

- Understand observed climate trends over India
- Understand seasonality and its changes.
- Understand how the distributions of climate variables have changed over the past years.
- Compute climate extremes indices and understand how extreme events have changed over the past years.

- Source: IMD gridded observation data
- Variables: Tavg, Tmax, Tmin, Rainfall
- Tavg, Tmax, Tmin: 1°×1° daily (1951–2023)
- Rainfall: 1°×1° daily (1901–2023)

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How the Session Will Work

We have shared **Google Drive folder** containing **dataset and Python notebook.**

1. **Add a shortcut** of the shared folder to your own Google Drive.
2. Open new notebook in Google Colab.
3. **Follow the provided notebook** for codes.
4. **Work on your own notebook.**

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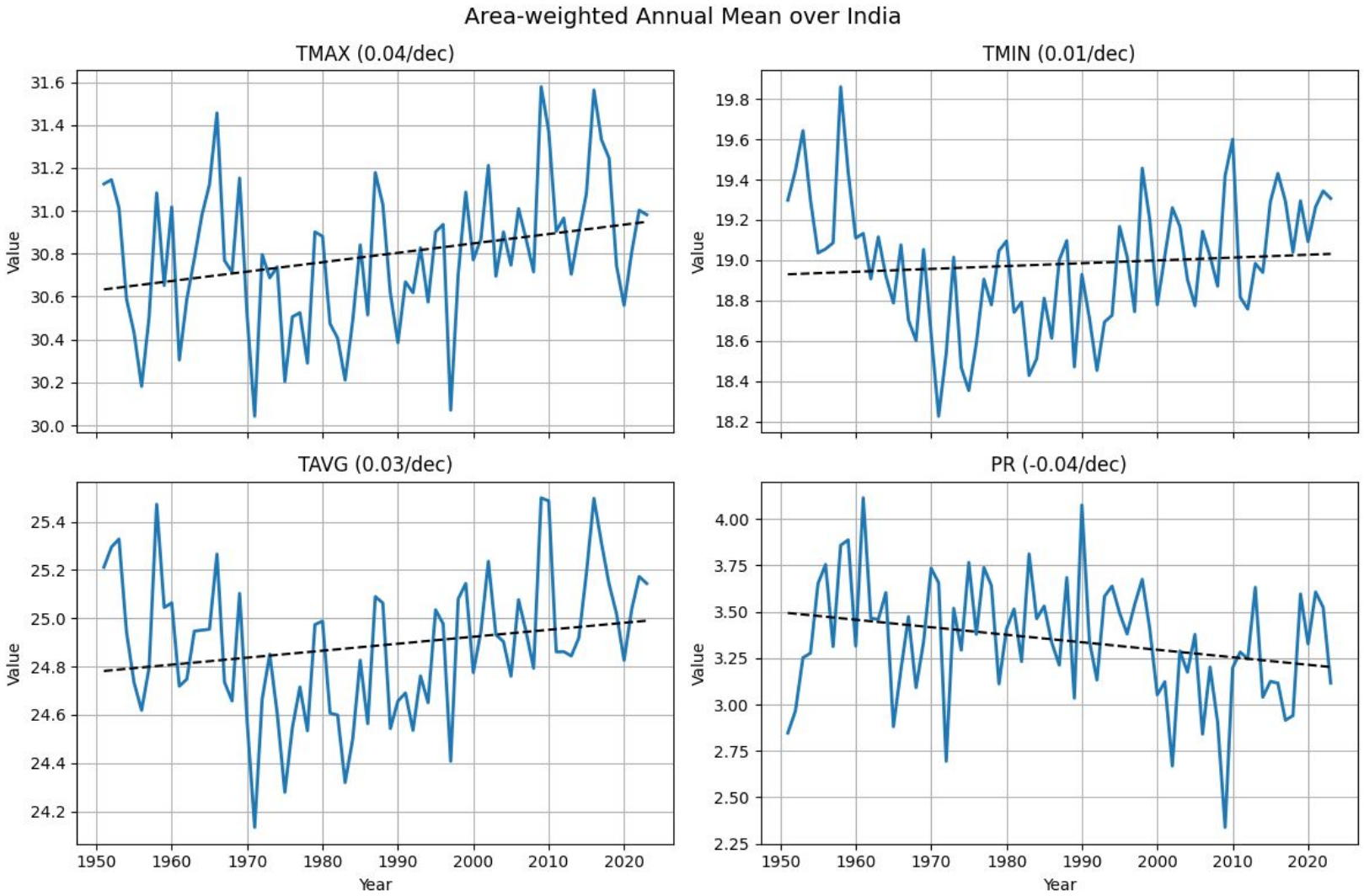
Hands-on Structure

- E1: Plot Annual Mean Time Series of the given climate variables and estimate the trends over India.
- E2: Plot the seasonal cycle of the temperature and precipitation over india over 2 different periods (e.g. 1951-1980 and 2005-2023)
- E3: Plot the Distribution of climate variables for two different periods.
- E4: Calculate the extreme indices and plot the time series of the extremes indices over India.

Exercise 1: Plot time series of tavg, tmin, tmax, pr and estimate trend

- How has the **annual mean Tmin, Tmax, Tavg and precipitation** changed over India from 1951–present?
- How different the trends in Tmax (Day time) and Tmin temperatures (night time) are?
- Do the trends remain same if you compute them for different periods? Comment on how the trends vary over different periods for different variables.

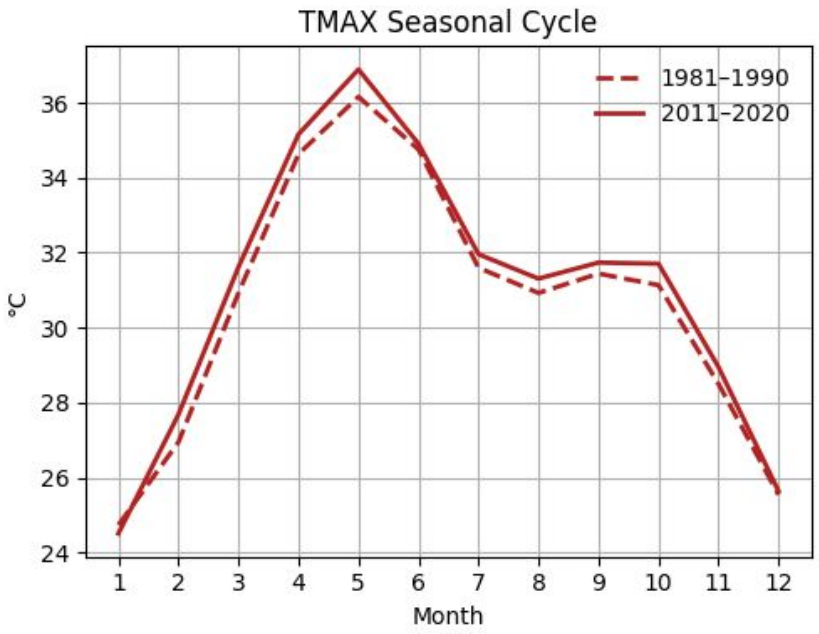
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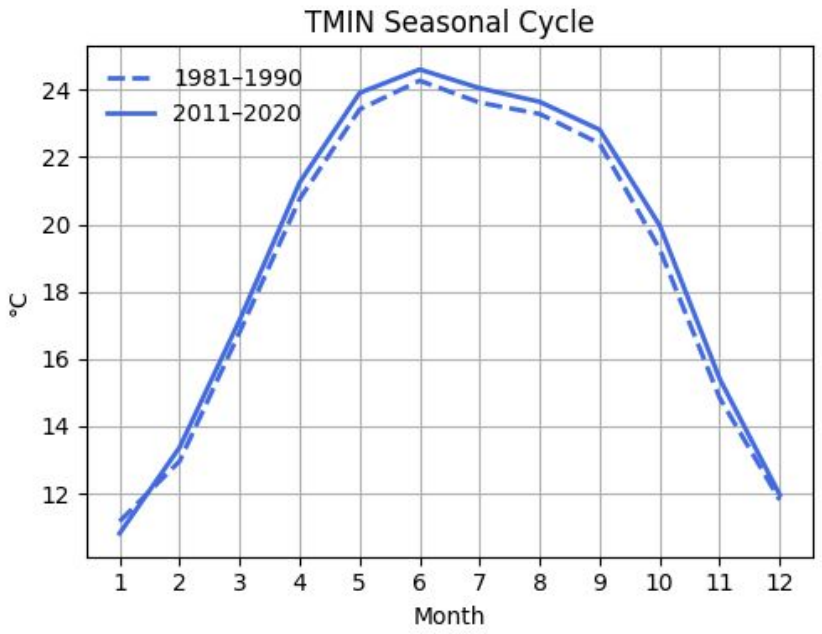


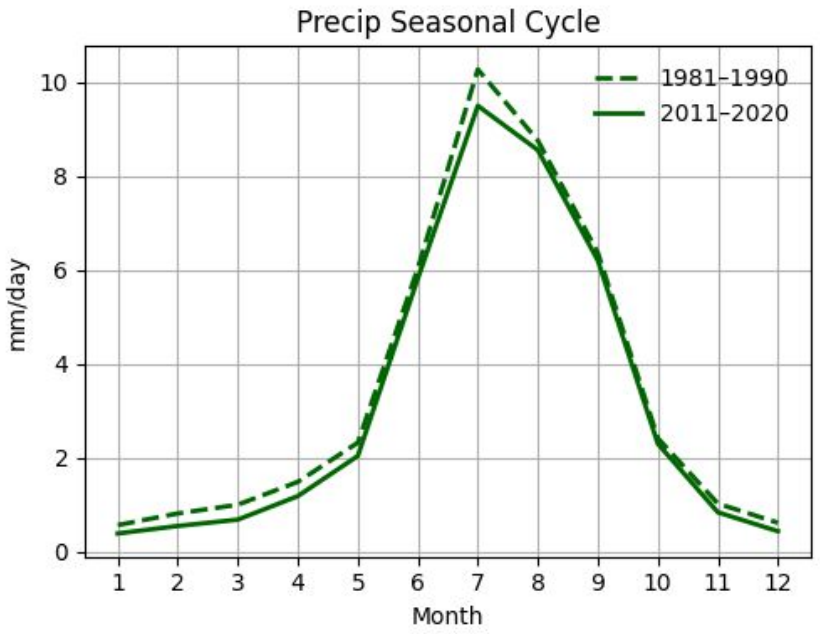
Exercise 2: Plot the monthly climatology to visualize the seasonal variability in the temperature and precipitation over india for 2 different periods.

- How has the **seasonal cycle (monthly climatology)** of Tmax, Tmin, and Pr changed between an earlier period and a recent period?
- Comment on the changes in monsoon precipitation, Daytime temperature during sumer and night time temperature in winter.

Exercise 2: Plot the monthly climatology to visualize the seasonal variability in the temperature and precipitation over india over 2 different periods.



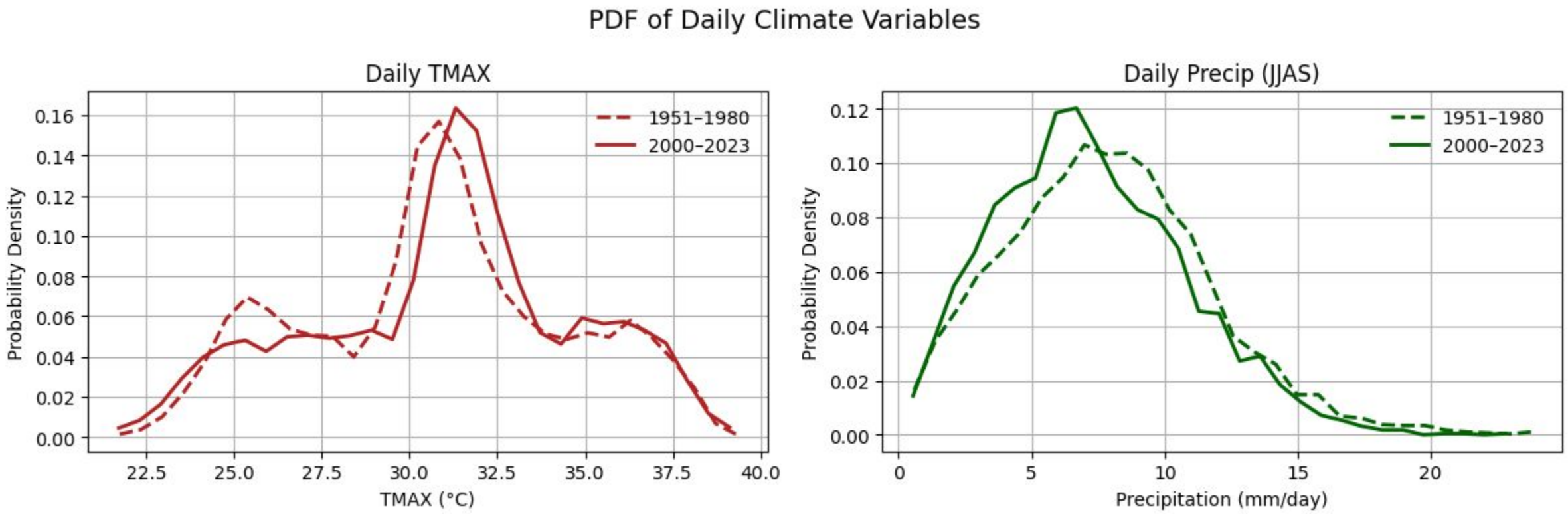




Exercise 3: Plot probability distribution of daily Tmin, Tmax, Tavg and Pr over two different periods

- How have the **probability distributions** of temperature and precipitation changed over time?
- Comment on the shift of the distribution in terms of mean and the extremes

Exercise 3: Plot probability distribution of daily Tmin, Tmax, Tavg, and PR over two different periods



Exercise 4: Compute climate extremes indices and plot the time series and trends of the Extreme indices.

- How does the extremes are changing over India?
- Are hot extremes increasing faster than the mean?
- Do extreme rainfall trends match mean rainfall trends? Comment on Mean vs Extreme rainfall changes?
- Check the same but for the lower tail extremes of Tmin (TNx) and compare (TXx and TNx)

Extreme Indices Definition

TX_x, *Yearly maximum value of daily maximum temperature*:
$TX_x = \max(TX)$

TNx Yearly maximum value of daily minimum temperature °C Minimum temperature

R95p *Annual total Extreme PRC: When RR > 95p*. Let RR_{wj} be the daily precipitation amount on a wet day *w (RR ≥ 1.0mm)* in year and let $RR_{wn}95$ be the 95th percentile of precipitation on wet days in the 1961-1990 period. If *W* represents the number of wet days in the year, then:

$$R95p_j = \sum_{w=1}^{W} RR_{wj} \text{ where } RR_{wj} > RR_{wn}95$$

How Extreme Indices are Calculated

HOW

For each grid:

- **Find the 95th percentile rainfall from 1961–1990**
- **Count how many days exceed the threshold each year**
- **Or sum how much rain falls above it.**

After finding extremes at each location, we average them to get an India value.

We do NOT average rainfall first and then find extremes.

WHY?

Example:

Grid A = 120 mm

Grid B = 5 mm

Regional mean = 62.5 mm

→ Looks non-extreme

- But one grid had an extreme event.

Exercise 4: Compute climate extremes indices and plot the time series and trends of the Extreme indices.

